

Predictive Maintenance Algorithm Based on Machine Learning for Industrial Asset

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Abstract—This article proposes a predictive maintenance algorithm based on machine learning to predict the remaining useful life of industrial assets. The synthetic dataset N-CMAPSS was used, which contains a performance degradation dataset until the presence of failure of an aircraft fleet under real flight conditions is detected. The principal element of maintenance focuses on the predictability of the remaining useful life; predictive models need performance information of an asset from the beginning to failure. [1]. The approach considers the data analysis to understand the data behavior. Monotonicity and principal component analysis are applied in the variable selection. Furthermore, convolutional neural networks are integrated to predict the remaining useful life, resulting in a 10.91 mean of RSME. The "DS01" dataset was used for training; six engines were used for the training dataset and the remaining four for the test dataset.

I. INTRODUCTION

Currently, the proper functioning of assets is a determining factor in a company's profitability and global competitiveness, being a critical element to reducing operating costs and minimizing downtime.

Particularly in the aerospace industry, it is considered that 28.6% of fatal accidents are related to maintenance. Accidents occur because of latent failures, with 30% caused by to engine and equipment stoppages [2].

The foregoing shows that the importance of adequate maintenance application is necessary for evaluating all elements that suffers wear, breakdowns and any other factors that affect the quality process.

In maintenance there are different strategies. For example, corrective maintenance consists of the immediate failure solution, and preventive maintenance concerns the evaluation and periodic replacement of asset parts. Finally, predictive maintenance focuses on data analysis to predict the best moment for applying maintenance [3]. Moreover, with Industry 4.0, prescriptive maintenance has been developed, providing practical advice for decision-making by, improving and optimizing future maintenance applications [4].

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In addition, with the emergence of industry 4.0, research regarding fast and accurate methodologies for remaining useful life predictions became important. For this reason, the need to understand the data through different techniques emerged.

Machine learning is a discipline focused on data analysis and learning. Some advantages are trend prediction, error minimization, process automation, and massive data storage. The results depend on quality and quantity data. Using machine learning to estimate the remaining lifespan allows modeling by extracting relevant patterns from the data without knowing the internal components [5]. The increasing availability of data has led different authors to use different methods for RUL prediction.

Specifically, [6] developed an algorithm to predict failures of a centrifugal pump with linear interpolation to complete missing data using SVM and MLP to predict the remaining useful life and obtain better results in MLP with 98.2% accuracy. In [7] spectral kurtosis and area-SK applied SVR are used to develop an RUL predictor for high-speed bearings obtaining better estimates using spectral kurtosis, and [8] used an SVM-based similarity approach for remaining life estimation with the CMAPSS dataset.

Some work has been done with neural networks. For example, in [9] feature maps were built for four different datasets using a training model based on CNN and validating it through k-fold, which obtained 12.16, 20.85, 12.01, and 24.97 RMSE for each dataset, respectively. On the other hand [10], the engine prediction process is simplified by using CNN sensors combined with a feature extractor, obtaining an average RSME of 19.53.

II. SIMULATION SYSTEM OF THE AIR OPERATION OF AN AIRPLANE ENGINE

The basic requirement for building predictive models is to have a run-to-failure dataset.

The C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) simulation system is an aircraft engine simulation under real flight conditions. Fig 1 represents the detailed diagram of the motor with the numbers of each variable as explained in the C-MAPSS model document [1]. The engine's thermodynamic model includes the ability to operate at altitudes from sea level to 40,000 feet, a Mach index of 0 to 0.90, and sea surface temperatures of -60°F to 103°F.

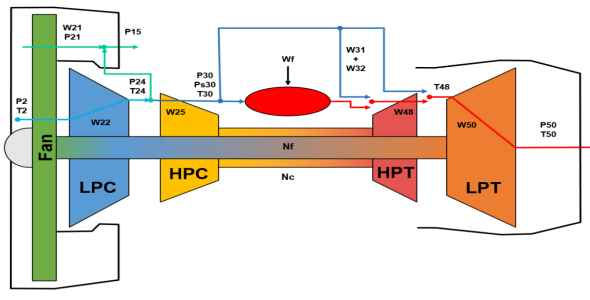


Fig. 1. Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) [1].

The simulator incorporates a system of non-linear equations. The input variables of the system model are divided into two parts, the operating state of the scenario descriptor W and the unobservable state parameters Θ .

The output refers to the measured physical properties estimated X_s and the unobservable properties X_v .

The system of nonlinear equations is described as:

$$[X_s^t, X_v^t] = F(W^t, \Theta^t) \quad (1)$$

The process of generating the dataset based on real flight data is obtained as shown in [1] and described below.

- 1) Record and define aircraft flight data
- 2) Enter degradation for each engine component
- 3) Flight parameters are simulated and wear is recorded
- 4) Health status is evaluated and it remains flying with the imposed degradation until it reaches 0.
- 5) Add disturbances to the sensor, the disturbances are included in the simulation data to explain the variation of the real readings

A. NEW-CMAPSS DATASET

The N-CMAPSS dataset used was developed from C-MAPSS. The dataset provides simulated data from a set of commercial aircraft engines with imposed performance degradation under real flight conditions. Moreover, the initial performance states are unknown.

At present, the N-CMAPSS set encompasses a total of eight data subsets of 128 units with seven different types of faults that alter the flow(F) and efficiency(E) of the engine's rotating components: the pressure of the engine compressor (LPC/HPC), the pressure in the turbine (LPT/HPT) and the fan.

Fig. 2 contains the data distribution for each of the eight data subsets, specifying the name of the data subset, the failure types, and the amount of data.

This work uses the DS01 subset that considers a failure mode. For each time series, a correct initial operation is considered based on the real flight conditions and in lapses of time, it begins to gradually degrade until a fault occurs.

The set contains 47 columns that represent the total input variables included for the prediction model. Each column includes data about the aircraft engine: operating conditions of the engine and sensor measurements.

Name	# Units	Flight Classes	Failure Modes	Fan		LPC		HPC		HPT		LPT		Size
				E	F	E	F	E	F	E	F	E	F	
DS01	10	1, 2, 3	1							✓				7.6 M
DS02	9	1, 2, 3	2							✓		✓	✓	6.5 M
DS03	15	1, 2, 3	1							✓		✓	✓	9.8 M
DS04	10	2, 3	1	✓	✓									10.0 M
DS05	10	1, 2, 3	1					✓	✓					6.9 M
DS06	10	1, 2, 3	1			✓	✓	✓	✓					6.8 M
DS07	10	1, 2, 3	1									✓	✓	7.2 M
DS08	54	1, 2, 3	1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	35.6 M

Fig. 2. Overview of datasets [1]

III. PROPOSED APPROACH

The work was developed using MATLAB R2021a software because of its forecasting and data analysis tools.

Fig. 3 illustrates the strategy applied for predicting remaining useful life. It focuses on three parts: dataset preprocessing, training, and maintenance.

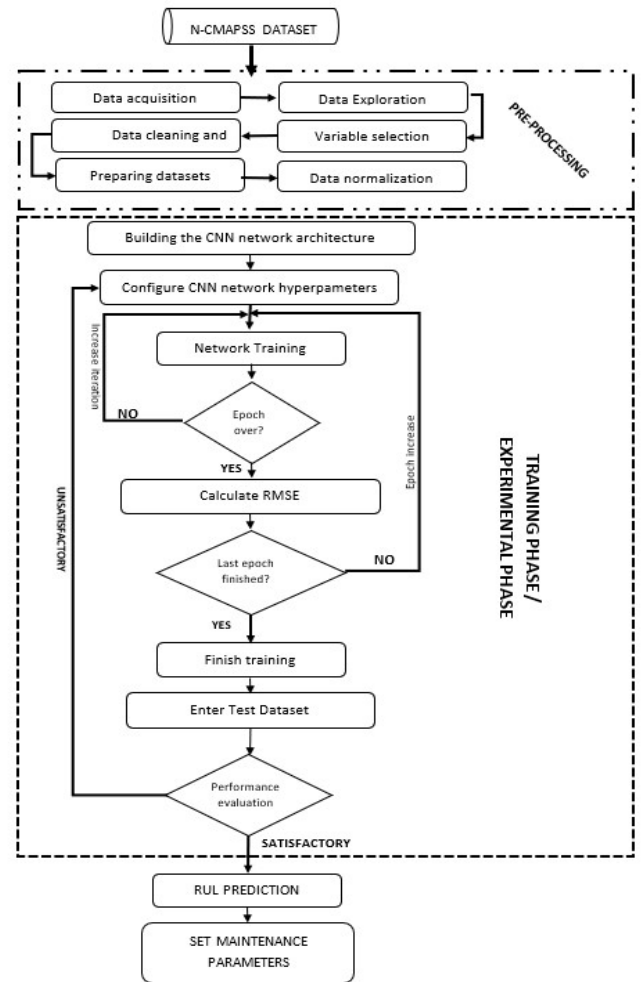


Fig. 3. Workflow of the proposed model for RUL prediction

A. DATASET PRE-PROCESSING

The pre-processing phase begins by extracting the data from the dataset and saving it in a table within MATLAB. The data are visualized to understand the behavior and structure.

In the adequate selection of variables for the investigation, each variable must describe one degradation part and the behavior of the variable for each instant of time.

To select the variables, we apply the monotonicity methodology. In the field of maintenance, as a system progressively approaches failure, a suitable condition indicator generally has a monotonic trend. If not, any feature with a non-monotonic trend is a less suitable indicator of condition [11].

The monotonicity of an information set is calculated by the difference between the positive and negative values, so it is considered a degradation process indicator [7].

The monotonicity equation is denoted as:

$$Monotonicity = \left| \frac{Poss(diff(x_i)) - Neg(diff(x_i))}{n - 1} \right| \quad (2)$$

- where n is the number of data measured over time.

For data filtering and smoothing, a moving average filter is applied, with a size corresponding to 4% of the total data of each engine.

To finish the pre-processing, the training and test datasets are prepared and the data are normalized to have predictors with zero mean and unit variance.

B. Training

The network architecture is designed in the training phase, and the network and training parameters are established (layers, epochs, batch sizes, etc.).

The network design was based on multiple convolutional layers. After each layer an activation function, a dropout layer, and finally fully connected layers are added.

Once the training is finished, the performance of the trained network is evaluated. If the result is not satisfactory, the parameters are modified until a satisfactory result is obtained.

C. Maintenance parameters

In the final part, with the estimated remaining useful life, maintenance is established, where it is recommended to determine thresholds for the asset's operation.

The maintenance application depends entirely on the researcher's knowledge about the studied asset. In this project, it is predicted based on the cycles or flights of each engine. Therefore, three thresholds were determined to establish the different states:

- 1) Operation with normal parameters (100% to 40% of RUL remaining).
- 2) Operation with some parameters of warning (39% to 16% of RUL remaining).
- 3) Operation of some parameters of alarm (15% to 0% of RUL remaining).

The metric used to evaluate the selected performance uses the root mean square error (RMSE) described in equations 3 and 4 [9].

$$\bar{x} = x_1 - x_2 \quad (3)$$

$$RMSE = \sqrt{\frac{1}{T} \sum_{i=1}^T \bar{X}} \quad (4)$$

- where x1 and x2 are the prediction accuracy and the true value of RUL
- $\bar{X}$ is the difference between x1 and x2
- And T is the number of samples

IV. RESULTS

For the development of the proposal, the DS01 dataset was used, which contains the information of 10 engines with a different number of flights and data recorded for each engine.

A. PRE-PROCESSING

To find the most appropriate condition indicators, the monotonic trend of the variables included in the dataset was quantified, obtaining the result that is shown in Fig. 4.

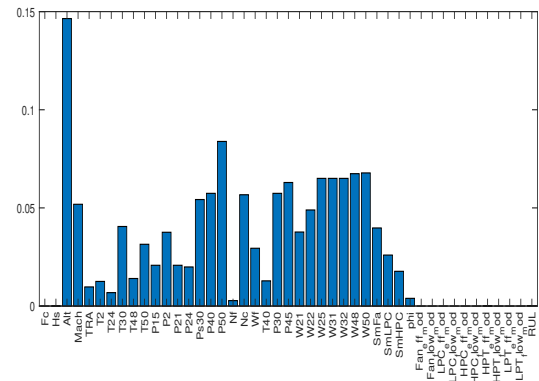


Fig. 4. Variable's monotonicity

By analyzing the histogram, a mean of 0.05 of the monotonic trend was obtained, so variables with values equal to or greater than the mean were selected. Therefore, these features are better at predicting the remaining wear life, as they are the best indicators of machine health.

The variables obtained are listed below:

- 1) Altitude (Altitude)
- 2) Mach speed (Mach)
- 3) Static pressure at HPC outlet (Ps30)
- 4) Total pressure at burner outlet (P40)
- 5) Total pressure at LPT outlet (P50)
- 6) Physical speed of the core (Nc)
- 7) Total pressure at HPC outlet (P30)
- 8) Total pressure at HPT outlet (P45)
- 9) Flow to HPC (W25)
- 10) HPT Coolant Bleed (W31)
- 11) HPT Coolant Bleed (W32)
- 12) Flow out HPT (W48)
- 13) Flow out of LPT (W50)

With the variables defined, a moving average filter is applied to smooth the data to get a better representation of them Fig. 5 shows the result of the application of the moving media filter, where the registered values of each variable (y-axis) at different moments (x-axis) are represented.

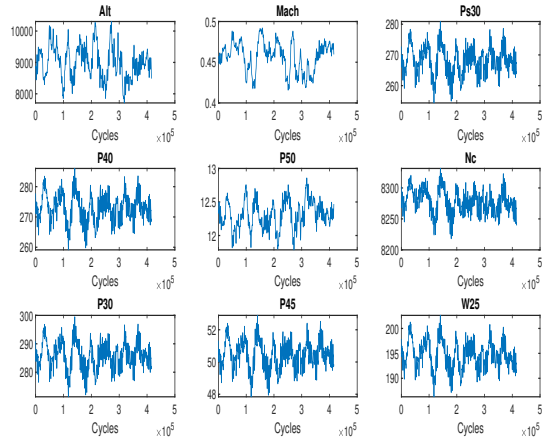


Fig. 5. Condition indicators with moving average filter

For the training dataset, the data of the first six engines were used, where only the selected variable's information was included. On the other hand, for the test dataset, the selected variable's information of the remaining four engines was used. Fig. 6 represents the training dataset.

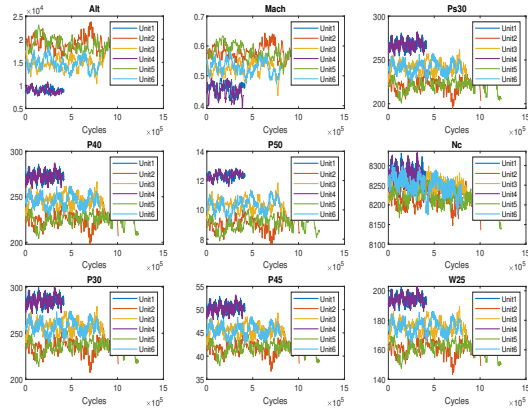


Fig. 6. Training dataset

To know the total data variance, the analysis of principal components was applied to them, it is important to mention that to apply PCA the data must be standardized.

The results when applying PCA indicate that, with the 13 variables selected, the first principal component explains 91% of the total data variance, while the second principal component explains around 6% of the variance. Therefore, the application of the first two principal components explains 97% of the total data.

Fig. 7 shows the results obtained by applying PCA, where the orange indicates the individual variance explained by each component, and the blue is the accumulated variance of the components.

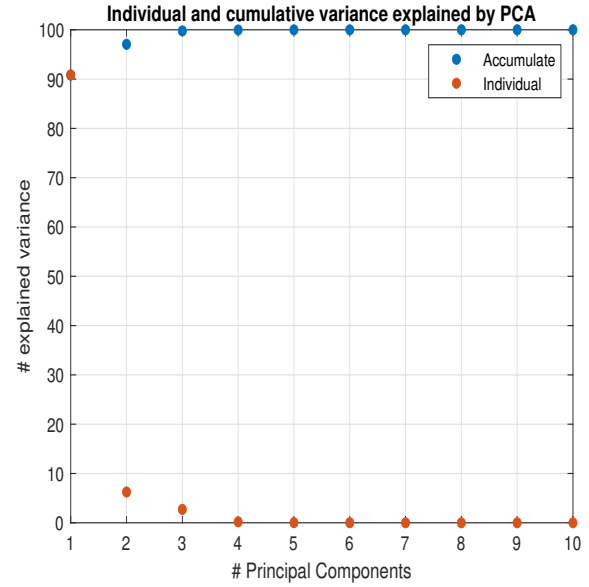


Fig. 7. Principal component analysis

B. Training

The training predictors are normalized to have zero mean and unit variance. Sequences with length of window size 250 are then generated from the time series data of each engine and a stride of 40. The corresponding response variable for each stream represents the RUL at the last timestamp of that stream.

For the input data, a specified size is expected as a row vector of integers, where the height, width, and several channels are described respectively. The convolutional neural network was developed, taking six engines from the dataset as training data.

The neural network architecture consists of four consecutive groups of CNN layers with one ReLU layer each, as well as a filter size and the number of filters as input arguments for the convolutional layer, followed by a completely connected layer of 100 hidden units and a layer with a dropout probability of 0.5 (Fig. 8).

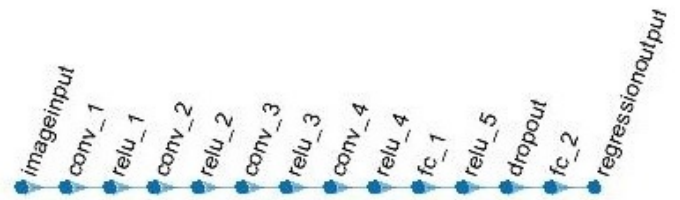


Fig. 8. Convolutional neural network architecture

The performance of the proposal was evaluated with the training data, obtaining an RMSE mean of 8.96, as shown in Fig 9.

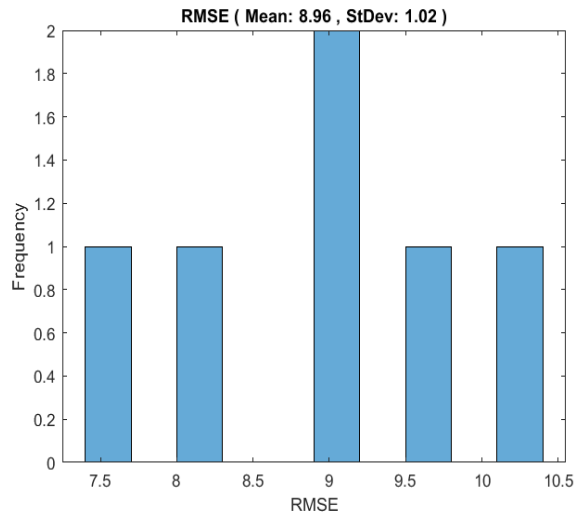


Fig. 9. RMSE of data training

In the same way, the performance of the test data were evaluated, where an RMSE with a mean of 10.91 was obtained, as seen in Figure 10..

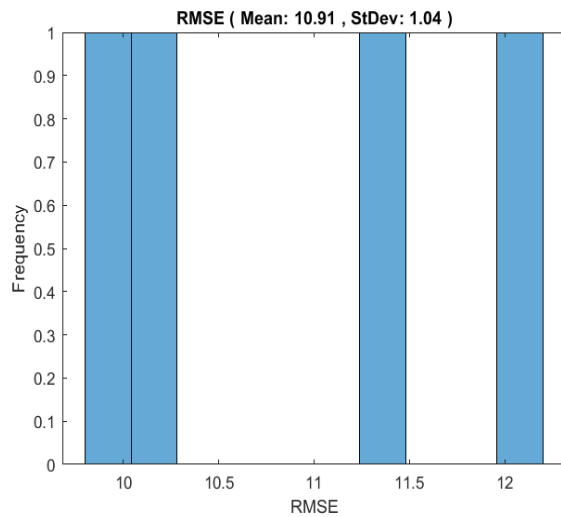


Fig. 10. RMSE of data training

For the training, 20 epochs with a mini-lot size of 512 were used. With the trained network, the remaining useful life is estimated.

In Figure 11, it is observed that the RUL has a linear degradation because this is the data provided by the dataset. The training and prediction are satisfactory since the RUL of the engine is predicted well accurately, having as an index a total of 87 flights referring to the RUL and the cycles as instants of time in which the data were collected.

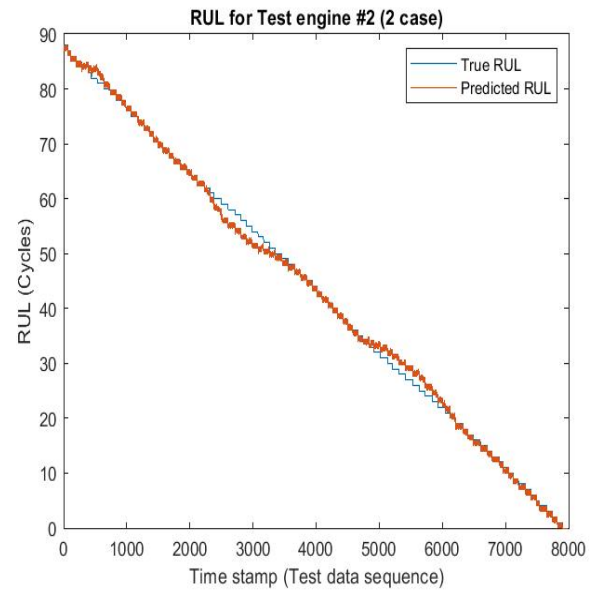


Fig. 11. Motor 2 Remaining Useful Life

C. Maintenance

According to the results obtained when predicting the remaining useful life, the operating thresholds are established to apply maintenance. In the second engine, the remaining life forecast counts a total of 87 flights. Green denotes the asset's normal operation that represents 87 to 37 flights remaining, and the yellow is the warning threshold that represents 36 to 14 flights remaining. It is recommended to apply maintenance before the red threshold entry, which indicates the alarm signal from a value of 13 to 0 of the remaining flights (Fig. 12).

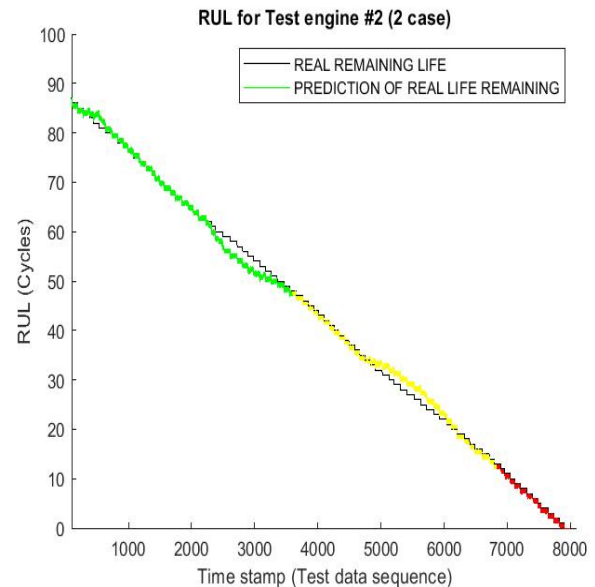


Fig. 12. Predictive maintenance

V. CONCLUSIONS

Engine wear was obtained from the information extracted by the sensors of the DS01 subset of the N-CMAPSS.

Based on the results, it is shown that the algorithm of the developed proposal responds by applying convolutional neural networks and obtaining a satisfactory RMSE mean that indicates that the network works, which allows the prediction of the remaining useful life based on the data.

It is important to mention that the results, when applied to a real case, could indicate variations because the study was performed with the new N-CMAPSS dataset.

For predictive maintenance this type of work is important because when applied correctly, the useful life of each component is used, reducing costs for part's replacement. In addition, it reduces the presence of failures and in turn, increases the safety and asset reliability.

VI. FUTURE WORK

In areas like maintenance and machine learning there are different approaches to implement. During the development of this work, some open ideas have emerged that it is hoped can be applied in future works.

Below are some ideas for future work that may be developed as a result of this work. In addition, some specific approaches are recommended to support and improve the proposal suggested.

Some possible research to develop:

- 1) Analyze how the engine flight classes influence, in this way a different prediction of the remaining useful life could be obtained.
- 2) Apply the proposed approach to other industrial assets where real operating data are available.
- 3) Develop the prescriptive maintenance approach to the data set.

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